

Does Seeing Diagonal Neighbors Help Preserve a Letter Z?

Moore-neighborhood context experiment · matched fixed seeds

Only statistical choice weighting changes. Horizontal/vertical compatibility, WFC propagation, and SAT CNF hard constraints remain exactly unchanged.

4-neighbor context

von Neumann = rook moves

N

W

x

E

S

8-neighbor context

Moore = king moves

NW

N

NE

W

x

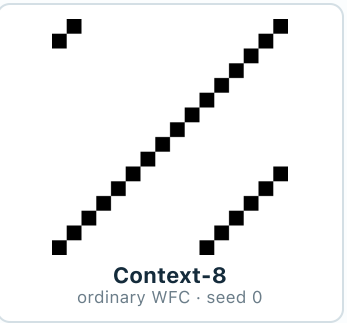
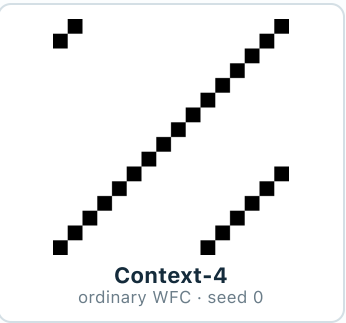
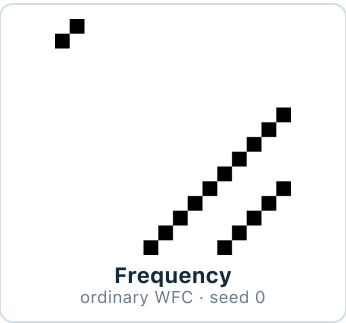
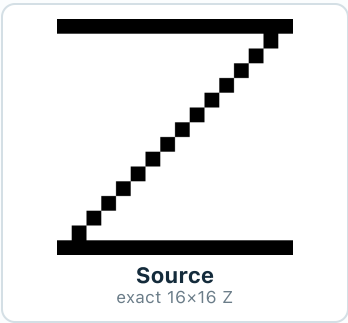
E

SW

S

SE

Main question: Does seeing diagonal neighbors help preserve the Z's diagonal?



Visually: no clear Moore win

The representative outputs preserve diagonal fragments, but none reconstructs a complete Z. Context-8 seed 0 is especially sparse.

Numerically: a small average gain

Across 20 matched seeds, Context-8 slightly improves diagonal and tagged local-8 resemblance versus Context-4. Smaller is better.

LETTER-Z · MEAN DIFFERENCE	HORIZONTAL	VERTICAL	DIAGONAL SE	DIAGONAL SW	TAGGED LOCAL-8	SUCCESS
Ordinary · Frequency	.1579	.0810	.0892	.1475	.1189	18/20
Ordinary · Context-4	.1526	.0700	.0764	.1438	.1107	18/20
Ordinary · Context-8	.1507	.0700	.0764	.1432	.1101	18/20
SAT · Frequency	.1726	.0989	.1061	.1537	.1329	20/20
SAT · Context-4	.1621	.0858	.0917	.1473	.1218	20/20
SAT · Context-8	.1641	.0823	.0882	.1496	.1211	20/20

Means use successful runs. Letter-Z fallback was 0% for both context methods. SAT conflicts: Frequency 177 · Context-4 167 · Context-8 9. Ordinary failures occurred on the same seeds (2 and 15) for all three weighting methods.

Moore Context Has Image-Dependent Effects

Representative seed-0 SAT images; quantitative findings use five matched seeds per benchmark.



SOURCE



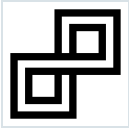
CONTEXT-4



CONTEXT-8

Cat · mixed

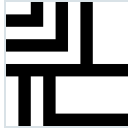
Moore improves vertical and both diagonal means for ordinary WFC, but Context-4 has the better SAT summary. SAT fallback rises from 15.6% to 28.2%.



SOURCE



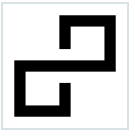
CONTEXT-4



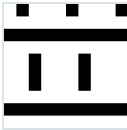
CONTEXT-8

Knot · Moore improves the measured means

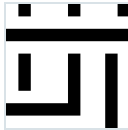
Context-8 is lower on horizontal, vertical, both diagonals, and the tagged local-8 score for both engines. Fallback rises to 4.0% ordinary and 11.3% SAT.



SOURCE



CONTEXT-4



CONTEXT-8

Simple Knot · Moore is worse

Context-8 degrades every reported mean relative to Context-4. SAT fallback rises from 6.6% to 41.6%, with conflicts increasing from 11 to 34.

What the experiment supports

- Moore context is not consistently superior; its effect depends on the image.
- Letter-Z diagonal statistics improve slightly, without a clear visual Z reconstruction.
- The larger context often increases unseen-context fallback, though not on letter-Z.
- Stick's Context-4 result remains stronger on horizontal/vertical resemblance; Moore is not an upgrade there.
- Conflict-free Stick outputs match between ordinary WFC and SAT. Under conflicts, Moore's CDCL interaction varies sharply by image.
- Moore looks worth targeted follow-up on diagonal-rich inputs, not adoption as a universal default.

Modest matched samples: 20 seeds for Stick and letter-Z; 5 for Cat, Knot, and Simple Knot. No significance tests.